

PRACTICALWEEK10

TASK1: Investigate inode structures using commands such as `ls -li`, `stat`, and `find`. Create hard links and symbolic links and explain their impact on inode allocation.

```
jhansi_priya@Jhansi-Priya: ~/   
jhansi_priya@Jhansi-Priya:~$ mkdir Practical9  
mkdir: cannot create directory 'Practical9': File exists  
jhansi_priya@Jhansi-Priya:~$ mkdir Practical10  
jhansi_priya@Jhansi-Priya:~$ cd Practical10  
-bash: cd: Practical10: No such file or directory  
jhansi_priya@Jhansi-Priya:~$ cd Practical10  
jhansi_priya@Jhansi-Priya:~/Practical10$ echo "Hello Ubuntu" > original.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ cat original.txt  
Hello Ubuntu  
jhansi_priya@Jhansi-Priya:~/Practical10$ cat original.txt  
Hello Ubuntu  
jhansi_priya@Jhansi-Priya:~/Practical10$ ls -li original.txt  
42112 original.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ stat original.txt  
File: original.txt  
Size: 13          Blocks: 8          IO Block: 4096   regular file  
Device: 8,48     Inode: 42112         Links: 1  
Access: (0644/-rw-r--r--)  Uid: ( 1000/jhansi_priya)   Gid: ( 1000/jhansi_priya)  
Access: 2026-09-22 03:59:21.317548819 +0000  
Modify: 2026-09-22 03:59:13.809548557 +0000  
Change: 2026-09-22 03:59:13.809548557 +0000  
Birth: 2026-09-22 03:59:13.809548557 +0000  
jhansi_priya@Jhansi-Priya:~/Practical10$ ln original.txt hardlink.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ ls -li original.txt hardlink.txt  
42112 -rw-r--r--  2 jhansi_priya jhansi_priya 13 Sep 22 03:59 hardlink.txt  
42112 -rw-r--r--  2 jhansi_priya jhansi_priya 13 Sep 22 03:59 original.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ echo "New line" >> hardlink.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ cat original.txt  
Hello Ubuntu  
New line  
jhansi_priya@Jhansi-Priya:~/Practical10$ ln -s original.txt symlink.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ ls -li original.txt hardlink.txt symlink.txt  
42112 -rw-r--r--  2 jhansi_priya jhansi_priya 22 Sep 22 04:00 hardlink.txt  
42112 -rw-r--r--  2 jhansi_priya jhansi_priya 22 Sep 22 04:00 original.txt  
2226 lrwxrwxrwx  1 jhansi_priya jhansi_priya 12 Sep 22 04:01 symlink.txt -> original.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ stat symlink.txt  
File: symlink.txt -> original.txt  
Size: 12          Blocks: 0          IO Block: 4096   symbolic link  
Device: 8,48     Inode: 2226         Links: 1  
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/jhansi_priya)   Gid: ( 1000/jhansi_priya)  
Access: 2026-09-22 04:01:09.759109042 +0000  
Modify: 2026-09-22 04:01:03.178270573 +0000  
Change: 2026-09-22 04:01:03.178270573 +0000  
Birth: 2026-09-22 04:01:03.178270573 +0000  
jhansi_priya@Jhansi-Priya:~/Practical10$ stat -L symlink.txt  
File: symlink.txt  
Size: 22          Blocks: 8          IO Block: 4096   regular file  
Device: 8,48     Inode: 42112         Links: 2
```

```
2226 lrwxrwxrwx  1 jhansi_priya jhansi_priya 12 Sep 22 04:01 symlink.txt -> original.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ stat symlink.txt  
File: symlink.txt -> original.txt  
Size: 12          Blocks: 0          IO Block: 4096   symbolic link  
Device: 8,48     Inode: 2226         Links: 1  
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/jhansi_priya)   Gid: ( 1000/jhansi_priya)  
Access: 2026-09-22 04:01:09.759109042 +0000  
Modify: 2026-09-22 04:01:03.178270573 +0000  
Change: 2026-09-22 04:01:03.178270573 +0000  
Birth: 2026-09-22 04:01:03.178270573 +0000  
jhansi_priya@Jhansi-Priya:~/Practical10$ stat -L symlink.txt  
File: symlink.txt  
Size: 22          Blocks: 8          IO Block: 4096   regular file  
Device: 8,48     Inode: 42112         Links: 2  
Access: (0644/-rw-r--r--)  Uid: ( 1000/jhansi_priya)   Gid: ( 1000/jhansi_priya)  
Access: 2026-09-22 04:00:53.028041899 +0000  
Modify: 2026-09-22 04:00:47.753603829 +0000  
Change: 2026-09-22 04:00:47.753603829 +0000  
Birth: 2026-09-22 03:59:13.809548557 +0000  
jhansi_priya@Jhansi-Priya:~/Practical10$ find .  
.  
./hardlink.txt  
./original.txt  
./symlink.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ ls -li original.txt  
42112 original.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ find . -inum 123456  
jhansi_priya@Jhansi-Priya:~/Practical10$ rm original.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ ls -li  
total 4  
42112 -rw-r--r--  1 jhansi_priya jhansi_priya 22 Sep 22 04:00 hardlink.txt  
2226 lrwxrwxrwx  1 jhansi_priya jhansi_priya 12 Sep 22 04:01 symlink.txt -> original.txt  
jhansi_priya@Jhansi-Priya:~/Practical10$ cat hardlink.txt  
Hello Ubuntu  
New line  
jhansi_priya@Jhansi-Priya:~/Practical10$ cat symlink.txt  
cat: symlink.txt: No such file or directory  
jhansi_priya@Jhansi-Priya:~/Practical10$ |
```

TASK2: Design a program using `mmap()` to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity.

```
GNU nano 7.2 task2.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main()
{
    const char *filename = "mmap_data.txt";
    const char *message = "Hello from mmap!";

    int fd;
    struct stat st;
    char *mapped;

    /* Open the file */
    fd = open(filename, O_RDWR | O_CREAT, 0666);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    /* Set the file size */
    if (ftruncate(fd, 100) == -1)
    {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    /* Get file information */
    if (fstat(fd, &st) == -1)
    {
        perror("fstat");
        close(fd);
    }
}
```

```
GNU nano 7.2 task2.c *
mapped = mmap(NULL,
               st.st_size,
               PROT_READ | PROT_WRITE,
               MAP_SHARED,
               fd,
               0);

if (mapped == MAP_FAILED)
{
    perror("mmap");
    close(fd);
    return 1;
}

/* Write data using memory mapping */
strcpy(mapped, message);

/* Make changes visible to the file */
if (msync(mapped, st.st_size, MS_SYNC) == -1)
{
    perror("msync");
}

printf("Data written using mmap(): %s\n", mapped);

/* Read data using memory mapping */
printf("Data read using mmap(): %s\n", mapped);

/* Unmap memory */
if (munmap(mapped, st.st_size) == -1)
{
    perror("munmap");
}

close(fd);

return 0;
}
```

```
jhansi_priya@Jhansi-Priya:~/Practical10$ nano task2.c

jhansi_priya@Jhansi-Priya:~/Practical10$ gcc task2.c -o task2
jhansi_priya@Jhansi-Priya:~/Practical10$ ls
hardlink.txt  symlink.txt  task2  task2.c
jhansi_priya@Jhansi-Priya:~/Practical10$ ./task2
File size: 100 bytes
Data written using mmap(): Hello from mmap!
Data read using mmap(): Hello from mmap!
jhansi_priya@Jhansi-Priya:~/Practical10$ ls -l task2.txt
ls: cannot access 'task2.txt': No such file or directory
jhansi_priya@Jhansi-Priya:~/Practical10$ ls -l task2_data.txt
ls: cannot access 'task2_data.txt': No such file or directory
jhansi_priya@Jhansi-Priya:~/Practical10$ ls -l task2
-rwxr-xr-x 1 jhansi_priya jhansi_priya 16392 Sep 22 04:05 task2
jhansi_priya@Jhansi-Priya:~/Practical10$ cat task2_data.txt
cat: task2_data.txt: No such file or directory
jhansi_priya@Jhansi-Priya:~/Practical10$
```

